



Lab Assignment 07 – Strings in Python



Instructions

1. Write each program in a separate Python file.
 2. Accept input from the user wherever required.
 3. Use **Strings, loops, and conditional statements** wherever appropriate.
 4. Display clear and meaningful output.
 5. Test each program with different inputs.
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1 Display Characters of a String

Write a Python program that accepts a word from the user and displays each character on a separate line using a loop.

Sample Output

```
Enter a word: Python

P
y
t
h
o
n
```

Hint: Use a `for` loop to iterate through the String.



2 Count Vowels in a String

Write a program that accepts a word or sentence and counts the number of **vowels**.

Consider both uppercase and lowercase vowels.

Sample Output

```
Enter text: Education

Number of Vowels = 5
```

Hint:

```
if letter in "aeiouAEIOU":
```

Use a variable such as `count` to keep track of the number of vowels.

3 Count a Particular Character

Write a Python program that:

- Accepts a word or sentence.
- Accepts a character to search for.
- Displays how many times that character occurs.

Sample Output

```
Enter text: banana
Enter character to search: a

Occurrences of a = 3
```

Hint: Loop through the String and compare each character with the character entered by the user.

4 Reverse a String

Write a Python program that accepts a word and displays it in **reverse order**.

Sample Output

```
Enter a word: Python

Original = Python
Reverse = nohtyP
```

Hint: Try solving it using a loop:

```
reverse = ""
```

Then build the reversed String one character at a time.

Challenge: After completing the loop-based solution, try using String slicing.

5 Palindrome Checker

A **palindrome** is a word that reads the same forwards and backwards.

Examples:

```
madam  
level  
radar  
civic
```

Write a program that accepts a word and checks whether it is a palindrome.

Sample Output 1

```
Enter a word: madam  
  
madam is a Palindrome.
```

Sample Output 2

```
Enter a word: python  
  
python is not a Palindrome.
```

Hint:

First reverse the word and then compare:

```
Original Word == Reversed Word
```

Use an **if-else** statement to display the result.



Bonus Question – Character Analyzer

Write a Python program that accepts a sentence and counts the number of:

- Uppercase letters
- Lowercase letters
- Digits
- Spaces

Sample Input

Enter text: Python Class 2026

Sample Output

```
Uppercase Letters = 2
Lowercase Letters = 9
Digits = 4
Spaces = 2
```

Hint: You can use:

```
letter.isupper()
letter.islower()
letter.isdigit()
letter.isspace()
```

Learning Outcomes

After completing this lab, students should be able to:

- Iterate through Strings using loops.
- Search and count characters.
- Combine Strings with conditional statements.
- Reverse Strings.
- Compare Strings.
- Identify palindromes.
- Use String methods such as `isupper()`, `islower()`, `isdigit()`, and `isspace()`.